Unit 3 - Session 2

SLO 2: Analyze the impact of noise and discretization on HT accuracy.

UG Level | Duration: 10 Minutes per Activity

Question:

"Why Did the Hough Transform Fail"

Q1. An image contains a clear straight line, but the Hough Transform fails to detect it. List one possible reason related to noise.

Answer: High noise may create false or weak edge points, making the actual line difficult to detect.

Q2. How can random edge noise affect the accumulator space even if the actual line is present?

Answer: Random edge noise produces scattered votes in the accumulator, creating many weak peaks and possibly hiding the true line peak.

Q3. The detected line appears shifted from the true line position. Which discretization issue can cause this?

Answer: Coarse discretization of the (ρ, θ) parameter space can place the detected line in a neighboring bin, causing a positional shift.

Q4. If multiple weak lines are detected instead of one strong line, what does this indicate about the discretization?

Answer: It indicates that the discretization is too fine, causing the votes for the same line to spread across several accumulator bins.

Q5. Consider two cases:
(a) High noise with fine discretization
(b) High noise with coarse discretization
Which case is more likely to produce a false dominant line? Why?

Answer: (b) High noise with coarse discretization is more likely to produce a false dominant line because many random votes can accumulate in the same large bin.

Q6. Why does increasing accumulator resolution not always improve Hough Transform accuracy?

Answer: Higher resolution can spread votes over more bins, making peaks weaker and increasing sensitivity to noise and small parameter errors.

Q7. Suggest one preprocessing step that can reduce HT errors caused by noise.

Answer: Apply Gaussian smoothing before edge detection to reduce noise and false edge points.